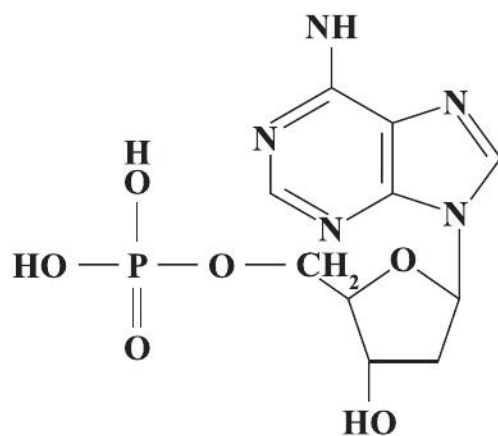
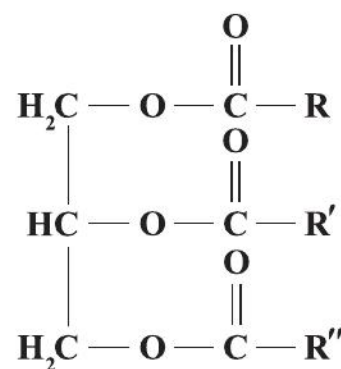


1. Which of the following molecules BEST represents the structure of a triglyceride?

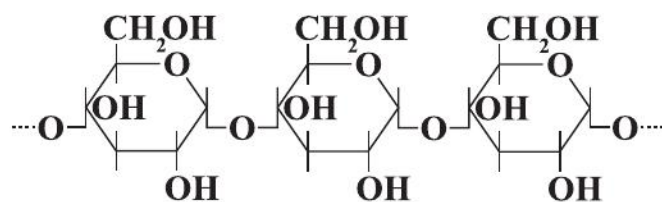
(A)



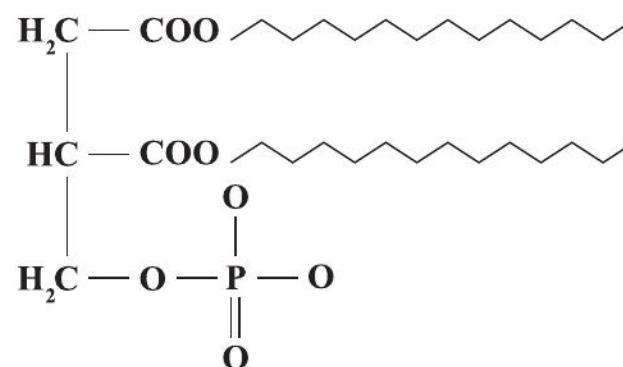
☐



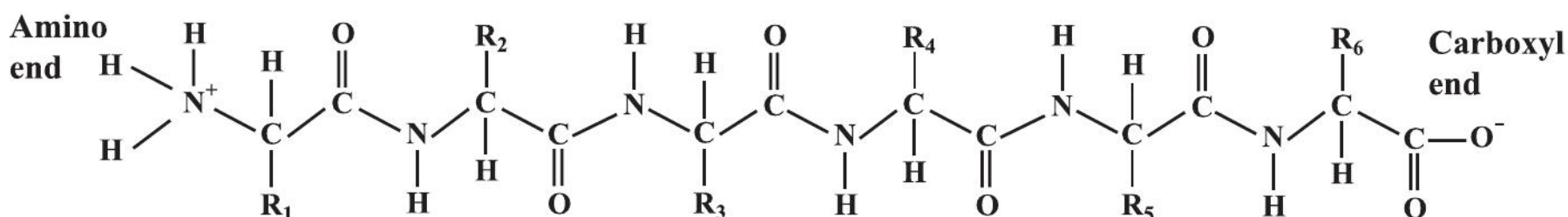
(B)



(D)



Item 2 refers to the following diagram of a protein structure.



2. During hydrolysis of the protein structure shown above, which of the following bonding interactions is broken?

- (A) Double
☐ (B) Peptide
 (C) Hydrogen
 (D) Hydrophobic

3. A diet high in triglycerides may lead to obesity. The feature of triglycerides which MOST likely contributes to this is that they

- (A) are insoluble in water
☐ (B) have many C-H bonds
 (C) contain glycerol molecules
 (D) have many -OH groups

4. Which of the following structures is NOT likely to be seen in an electron micrograph of a plant cell?

- ☐ (A) Centriole
 (B) Tonoplast
 (C) Ribosome
 (D) Golgi apparatus

5. What is the HIGHEST level of protein structure exhibited by haemoglobin?

- (A) Primary
- (B) Secondary
- (C) Tertiary
- ☒ (D) Quaternary

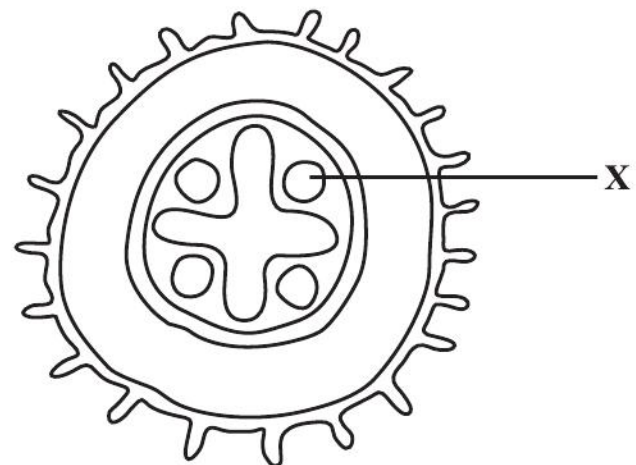
6. Which structure is found in BOTH a prokaryotic cell and an eukaryotic cell?

- ☒ (A) Ribosome
- (B) Chloroplast
- (C) Mitochondrion
- (D) Nuclear membrane

7. Collagen exhibits a secondary level of protein structure and organization because

- (A) it contains four polypeptide chains
- (B) each chain forms a helix held by covalent bonds
- ☒ (C) each chain forms a helix held by hydrogen bonds
- (D) it is made up of repeated units of three different amino acids

Item 8 refers to the following diagram which shows a transverse section of a plant structure.



8. The area labelled X represents

- (A) a cell
- ☒ (B) a tissue
- (C) an organ
- (D) a system

9. Bobby received a tropical freshwater fish for his birthday. While experimenting with the fish, he places it in seawater and it dies. Which of the following statements MOST likely explains the reason for its death?

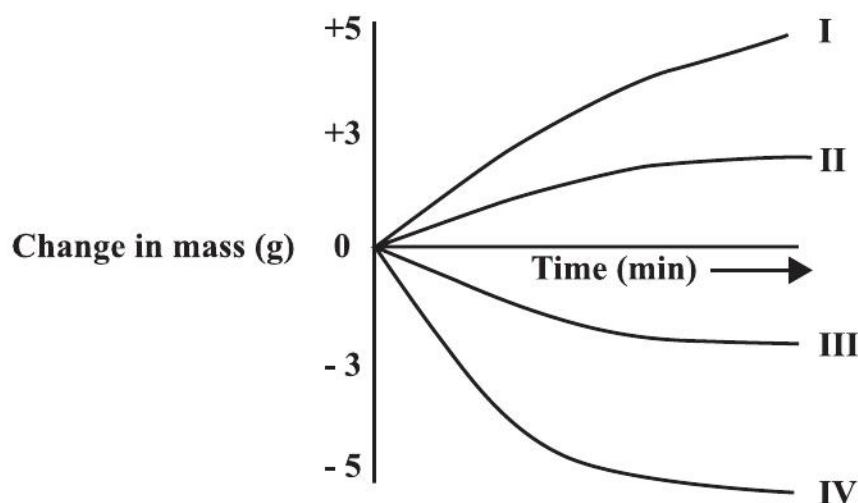
- (A) Water diffused into the fish.
- ☒ (B) Water diffused out of the fish.
- (C) Ions diffused from the fish to the seawater.
- (D) Less oxygen is in seawater than freshwater.

10. The following table shows three processes involved in the movement of substances across cell membranes. Which of the following rows correctly classifies each process as active and/or passive in nature?

	Diffusion	Endocytosis	Osmosis
(A)	Active	Active	Active
(B)	Active	Passive	Active
☒ (C)	Passive	Active	Passive
(D)	Passive	Passive	Passive

11. Four cylinders of equal mass are cut from a single potato. One cylinder is placed in distilled water and the others in 0.2 M, 0.4 M and 0.8 M sucrose solution, respectively. Each cylinder is weighed at half-hour intervals for six hours. The graph below shows the results obtained.

Which curve MOST likely represents the 0.4 M sucrose solution?



- (A) I
(B) II
☒ (C) III
(D) IV

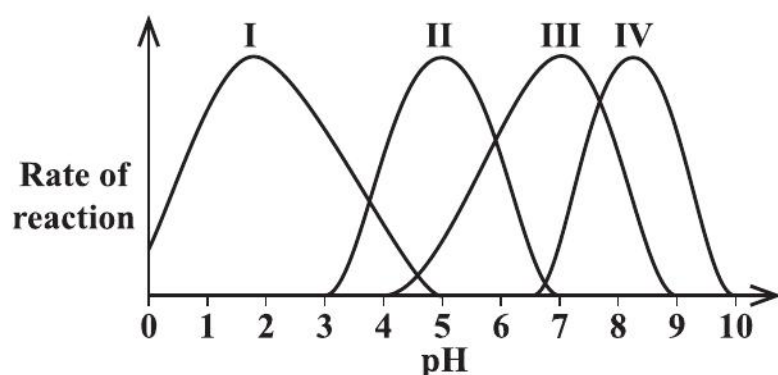
12. Which of the following correctly describes the importance of cholesterol in a cell membrane?

- (A) It interacts with fatty acid tails to exclude water.
(B) It forms channels in the membrane facilitating the entry and exit of polar molecules.
(C) It has branching carbohydrate chains which act as identity markers.
☒ (D) It enhances the flexibility and mechanical stability of the membrane.

13. Which of the following is the BEST example of anabolism?

- (A) Glycogen is converted to glucose.
(B) Peptide bonds are hydrolysed by peptidases.
☒ (C) Amino acids are joined by peptide bonds to form a polypeptide.
(D) Amylose and amylopectin combine to form starch molecules.

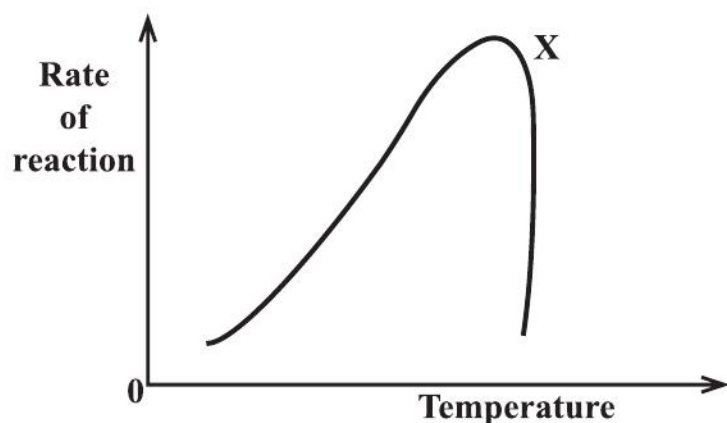
Item 14 refers to the following activity curve which show the effect of pH on the activity of enzymes.



14. Pepsin is an enzyme found in the human alimentary canal. Which of the activity curves above BEST represents the action of pepsin?

- ☐ (A) I
☐ (B) II
☐ (C) III
☐ (D) IV

Item 15 refers to the following graph which shows the progress of an enzyme-catalysed reaction with increasing temperature.

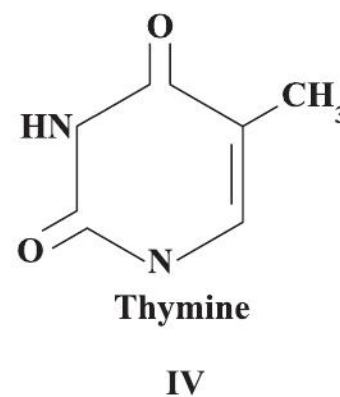
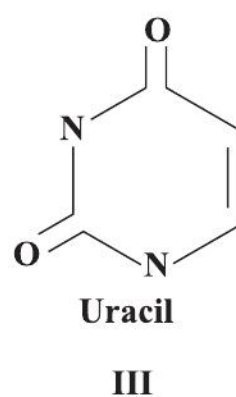
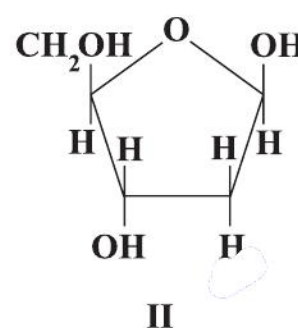
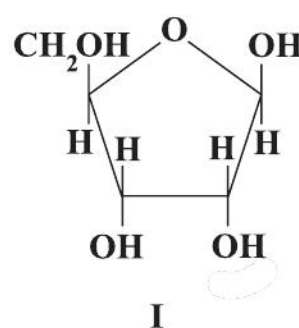


15. The decline in the rate at X is MOST likely because
- (A) peptide bonds are broken
 (B) the enzyme loses its secondary structure
☐ (C) the enzyme loses its tertiary structure
 (D) substrate concentration becomes too low

16. Which of the following is found in BOTH DNA and transfer RNA?

- (A) Uracil
 (B) Ribose
☐ (C) Sugar-phosphate backbone
 (D) Double helix structure

Item 17 refers to the following diagrams of molecules which are involved in the formation of nucleic acids.



17. Which TWO molecules when combined with a phosphate group form an RNA nucleotide?

- ☐ (A) I and III
☐ (B) I and IV
☐ (C) II and III
☐ (D) II and IV

Item 18 refers to the following table which shows the codes and the matching amino acids.

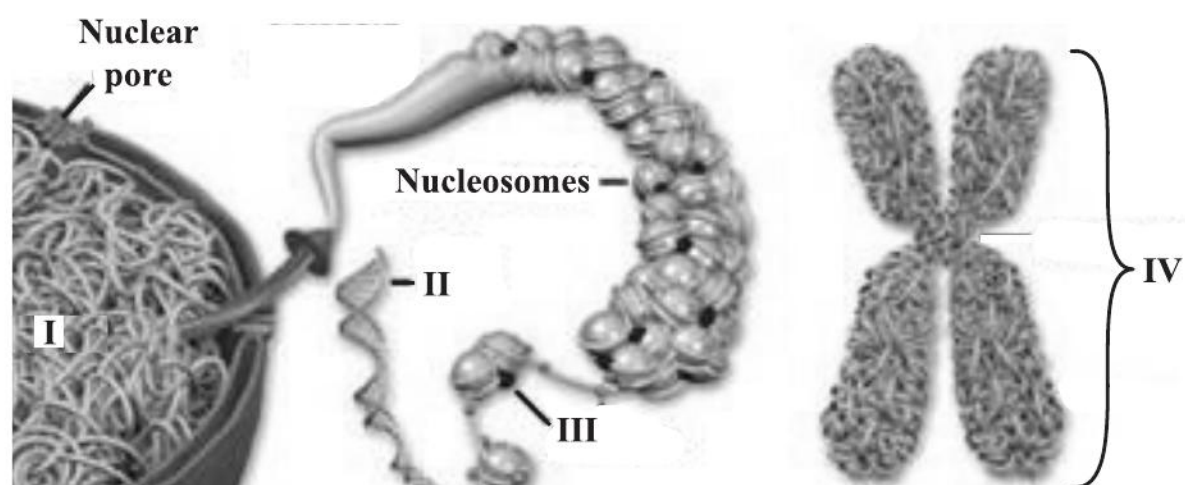
ACC–thr	UUA–leu	CUA–leu	CCG–pro
CGG–arg	CCA–pro	UAC–tyr	GGC–gly

18. Which of the following amino acid sequences would be produced from information coded on the DNA strand shown below?

ATGGCCAATGGCGAT

- (A) leu–pro–tyr–gly–arg
☐ tyr–arg–leu–pro–leu
 (C) pro–gly–leu–thr–arg
 (D) tyr–arg–gly–pro–leu

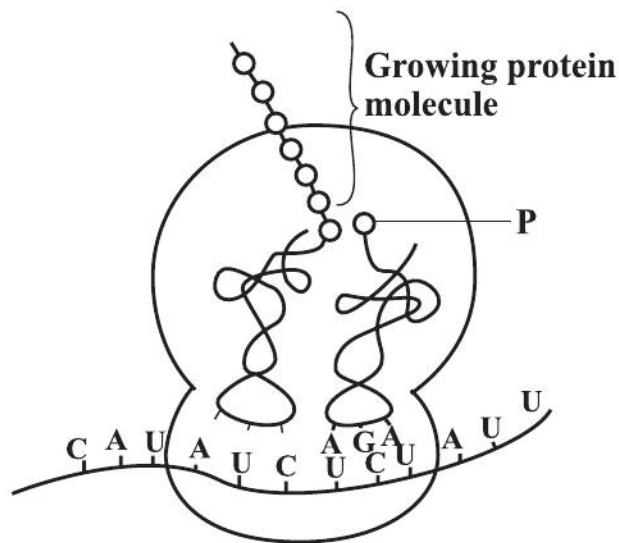
Item 19 refers to the following diagram (not drawn to scale) which shows nuclear material in a cell.



19. Which of the following rows matches each structure with its correct name?

	I	II	III	IV
(A)	Chromosome	DNA	Histone	Chromatid
(B)	Chromosome	Chromatin	DNA	Chromatid
☐	Chromatin	DNA	Histone	Chromosome
(D)	DNA	Histone	Chromatid	Chromosome

Item 20 refers to the following process of translation occurring at a ribosome in a cell that synthesizes enzymes.



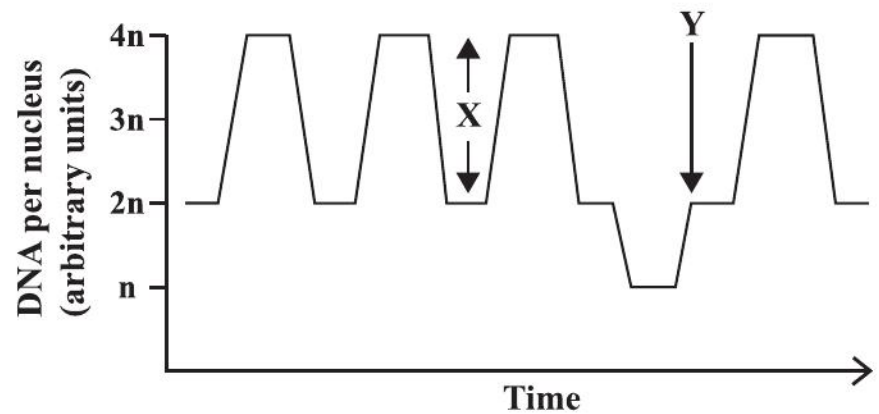
20. The structure, **P**, MOST likely represents

- (A) a protein
- (B) a ribosome
- (C) a nucleotide
- ☒ (D) an amino acid

21. Which of the following statements is NOT true of most homologous chromosomes?

- (A) They are the same length.
- ☒ (B) They contain the same alleles.
- (C) They contain the same number of genes.
- (D) They contain the same heterochromatin bands.

Item 22 refers to the following diagram which represents the relative amounts of DNA per nucleus during many cell divisions in an animal cell.



22. Which of the following processes are occurring at X and Y?

	X	Y
(A)	Fertilization	DNA synthesis
(B)	DNA synthesis	Separation of chromosomes
☒ (C)	DNA synthesis	Fertilization
(D)	Fertilization	Cytokinesis

Item 23 refers to the following information about fur colour in cats.

A particular breed of domestic cats can have either black, white or brown fur colour. The dominant allele, B, of one gene gives black fur and the recessive allele, b, gives brown fur.

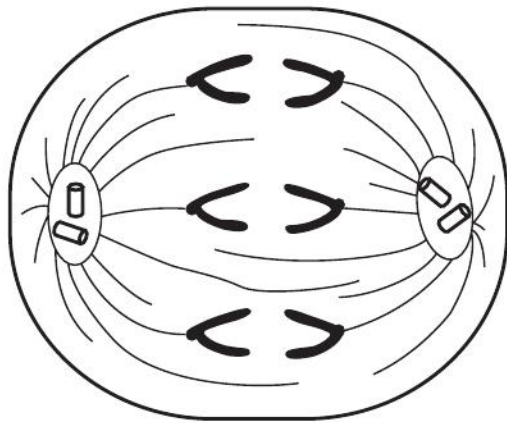
Many of the white cats carry a dominant allele, P, of a second gene which inhibits all pigment production. The recessive allele, p, has no effect on the colour of the fur.

The two different genes are not linked nor are they on the X chromosome.

23. This type of gene interaction is described as

- ☒ (A) epistasis
- (B) sex linkage
- (C) codominance
- (D) dominant/recessive

Item 24 refers to the following diagram showing one stage of mitotic cell division.



24. From the position and shape of the chromosomes, what stage of **mitosis** has been reached?

(A) Mid-metaphase
(B) Early telophase
☐ (C) Early anaphase
(D) Mid-prophase

25. Mutations may arise due to non-disjunction of chromosomes. In which of the following processes would such a mutation be transmitted to the offspring?

(A) During the division of cells in the cambium
(B) During the division of malignant cells in the skin
(C) After the fusion of a male nucleus with the polar nucleus in a plant
☐ (D) During the development of the secondary oocyte in an animal

26. A possible benefit of producing transgenic plants is

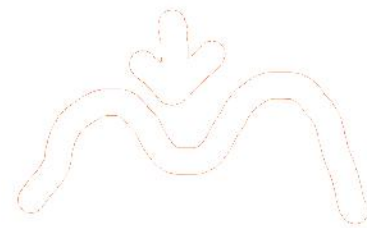
(A) determination of human genome
☐ (B) improved human nutrition
(C) increased use of pesticides
(D) production of clones

27. Which of the following events that occur during meiosis is MOST likely to result in chromosomal mutation?

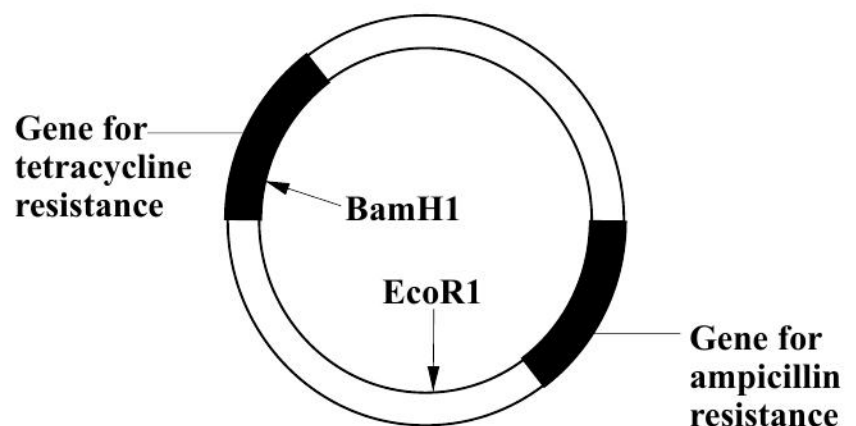
☐ (A) Non-disjunction
(B) DNA replication
(C) Homologous pairing
(D) Independent assortment

28. In populations of Pacific salmon, most fish are either very large or very small. This is because sperm from medium-sized fish seldom fertilize the eggs. The allele frequency of medium-sized fish is therefore very low. This MOST likely is an example of

(A) artificial selection
(B) stabilizing selection
(C) directional selection
☐ (D) disruptive selection



Item 29 refers to the following map of a plasmid showing the position of recognition sequences of two restriction enzymes.



29. In addition to the human insulin gene, which of the following enzymes would be required to produce a recombinant plasmid?

(A) ADPase
☐ (B) DNA ligase
(C) RNA polymerase
(D) Reverse transcriptase

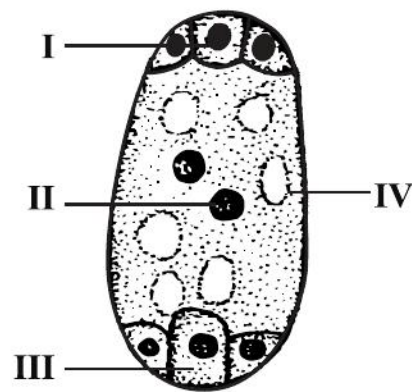
30. Which type of isolation plays the MOST significant role in allopatric speciation?

- ☐ (A) Geographical
- ☐ (B) Temporal
- ☐ (C) Ecological
- ☐ (D) Behavioural

31. Which of the following structures are formed during the development of pollen grains?

- ☐ (A) Antipodals
- ☐ (B) Synergids
- ☐ (C) Polar nuclei
- ☐ (D) Tube nuclei

Item 32 refers to the following diagram which shows the embryo sac of a flowering plant.



32. Which of the labelled structures develops into the endosperm?

- ☐ (A) I
- ☐ (B) II
- ☐ (C) III
- ☐ (D) IV

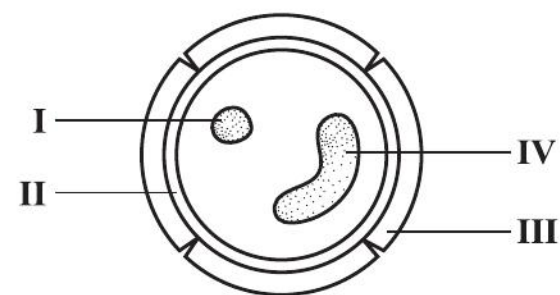
33. Which of the following terms is MOST appropriate to describe a plant in which the anthers of the flower mature before the stigma?

- ☐ (A) Protandry
- ☐ (B) Protogyny
- ☐ (C) Monoecious
- ☐ (D) Hermaphrodite

34. Which of the following statements is true about double fertilization in flowering plants?

- ☐ (A) Two sperm nuclei fuse with one egg cell.
- ☐ (B) Two sperm nuclei fuse with two egg cells.
- ☐ (C) One sperm nucleus fuses with both an egg cell and a polar nucleus.
- ☐ (D) Two sperm nuclei fuse with an egg cell and polar nuclei, respectively.

Item 35 refers to the following diagram showing a mature pollen grain.



35. Which of the labelled structures will give rise to two male gametes?

- ☐ (A) I
- ☐ (B) II
- ☐ (C) III
- ☐ (D) IV

36. Which of the following conditions does NOT promote cross-pollination?

- (A) Self-incompatibility
- ☒ (B) Cleistogamy
- (C) Protogyny
- (D) Protandry

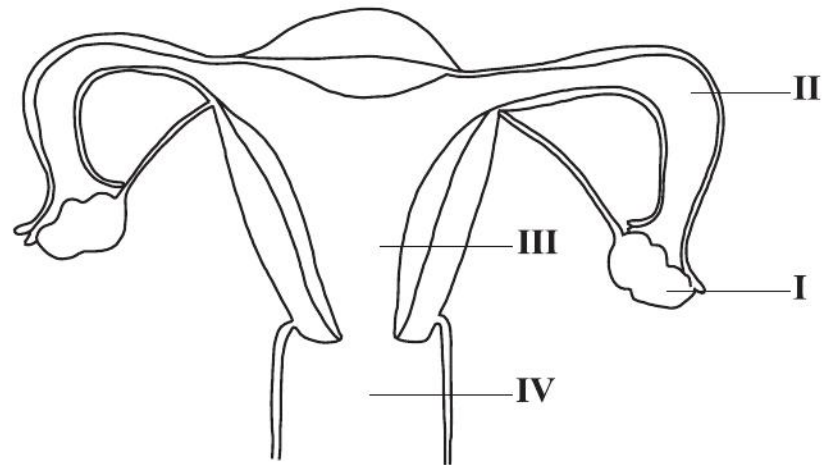
37. In plants, the ovary is BEST described as the

- (A) tip of a carpel
- (B) basic unit of the gynoecium
- ☒ (C) swollen base of a single carpel
- (D) female reproductive part of the flower

38. Which of the following statements identifies a difference between the secondary oocyte and the ovum?

- (A) The secondary oocyte is diploid while the ovum is haploid.
- (B) The secondary oocyte has two polar bodies while the ovum has one.
- (C) The secondary oocyte is surrounded by the zona pellucida while the ovum is not.
- ☒ (D) The secondary oocyte is at Metaphase II of meiosis while the ovum has completed the meiotic divisions.

Item 39 refers to the following diagram of the human female reproductive system.



39. At which of the regions labelled I, II, III and IV does fertilization take place?

- (A) I
- ☒ (B) II
- (C) III
- (D) IV

40. Human spermatozoa have to spend several hours in the female genital tract before they can fertilize an oocyte. During this time, the spermatozoa are activated in a process called

- ☒ (A) capacitation
- (B) ejaculation
- (C) conception
- (D) copulation

41. Which of the following events occurring during the ovarian cycle is a result of the negative feedback caused by an increase in oestrogen levels?

- (A) Decreased production of progesterone at the end of the secretory phase
- (B) Increased secretion of luteinizing hormone during the follicular phase
- (C) Increased secretion of gonadotrophin releasing hormone during the follicular phase
- ☒ (D) Decreased secretion of follicle-stimulating hormone after the release of the secondary oocyte

42. A sample of a woman's urine is tested at a laboratory and human chorionic gonadotrophin is detected. This indicates that

- (A) ovulation has occurred
- (B) menstruation has begun
- (C) oestrogen levels have decreased
- ☒ (D) implantation of an embryo has occurred

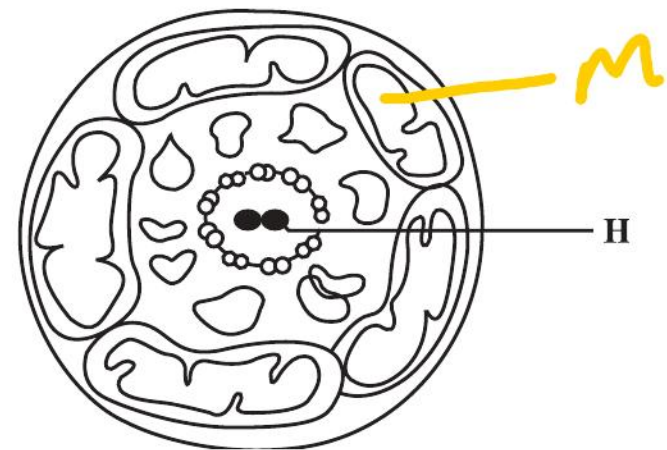
Items 43–44 refer to the following hormones.

- (A) Oestrogen
- (B) Progesterone
- (C) Luteinizing hormone
- (D) Follicle-stimulating hormone

43. Which hormone stimulates the development of the endometrium? **A**

44. Which hormone regulates the precise timing of ovulation? **C**

Item 45 refers to the following diagram showing a transverse section through the middle piece of a sperm cell.



45. The structure labelled H is

- (A) a nucleus
- (B) an acrosome
- ☒ (C) a microtubule
- (D) a mitochondrion